KRS Dam Rainfall Pipeline & MLOps Reproducibility Report

Branch: demo | **Commit:** 1b26c7e | **Date:** 2026-11-05

1. Gate Status Summary

Gate	Scope	Status	Notes
Gate 1	Unit Conversions	DONE	Validated via tests/test_unit_conversions.py
Gate 2	Storage Safeguards	DONE	Validated via tests/test_storage_analysis.py
Gate 3	Water Balance	DONE	Validated via tests/test_water_balance.py
Gate 4	Freshness & Inference	DONE	Validated via tests/test_inference_freshness.py
Gate 5	Rainfall Pipeline	BLOCKED_EXTERNAL_SOURCE	Raw data absent; no synthetic data fabricated
Gate 8	MLOps Reproducibility	PARTIAL_LOCAL_ONLY	Local DVC/MLflow verified; cloud remote unconfigured

2. Verification & Test Matrix

- **Layer A Core Tests:** 49 passed (100%)
- **Rainfall Schema & Anti-Leakage Tests:** 20 passed (100%)
- **Total Passing Tests:** 69 passing out of 69 executed

3. Key Reports Generated

- *reports/rainfall_source_blocker.md* (Provider & raw data gap)
- *reports/rainfall_data_audit.md* (Station proxies & calendar merge)
- *reports/rainfall_feature_contract.md* (t-1 cutoff & anti-leakage)
- *reports/paired_forecast_comparison.md/.csv* (Evaluation protocol)
- *reports/dvc_reproducibility.md* (DVC DAG & remote cache audit)
- *reports/mlflow_reproducibility.md* (MLflow isolated store audit)
- *reports/reproducibility_manifest.json* (Machine-readable audit)